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Note: Live quiz MCQs also repeated in Exam (6, 7)



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# MTH631

## Mid Term (Live Quiz)

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03184148783
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**1. The sequence sn(x)=a02+∑k=1n(akcos⁡kx+bksin⁡kx),x∈R,s_n(x) = \frac{a_0}{2} + \sum_{k=1}^n (a_k \cos kx + b_k \sin kx), x \in \mathbb{R}, converges on a set _____.**

- R ✓
- E
- O
- N

---

**2. Let ε\varepsilon be the set of all real-valued piecewise-defined periodic functions ff on the interval.**

- (0, 2π]
- [−2π, 2π]
- [−π, π] ✓
- [a, b]

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**3. The expression 1π∫−ππcos⁡(nx)cos⁡(kx)dx=\frac{1}{\pi} \int_{-\pi}^\pi \cos(nx) \cos(kx) dx = _____.**

- 1 if n≠kn \neq k
- 1 if n=kn = k ✓
- 0 if n=kn = k

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- ∞\infty if n≠kn \neq k

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**4. Let ff be defined by a power series with finite radius of convergence RR. If ∑n=0∞(−1)nanRn\sum_{n=0}^\infty (-1)^n a_n R^n diverges, then lim⁡x→(x0+R)+f(x)=∑n=0∞(−1)nanRn\lim_{x \to (x_0 + R)^+} f(x) = \sum_{n=0}^\infty (-1)^n a_n R^n.**

- True ✓
- False

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**5. Let Φ={φ1,φ2,…,φn,… }\Phi = \{\phi_1, \phi_2, \dots, \phi_n, \dots\} be an orthonormal basis of an infinite-dimensional inner product space χ\chi, and let f∈χf \in \chi. Then the infinite series _____ is called the Fourier series.**

- ∑k=1∞⟨f,φ0⟩φk(x)\sum_{k=1}^\infty \langle f, \phi_0 \rangle \phi_k(x)
- ∑k=1∞⟨f,φk⟩φk(x)\sum_{k=1}^\infty \langle f, \phi_k \rangle \phi_k(x) ✓
- ∑k=1∞⟨f,φk(x)⟩\sum_{k=1}^\infty \langle f, \phi_k(x) \rangle
- None of these

---

**6. A Fourier series expansion of a function is a representation of the function as a linear combination of sines and cosines as _____.**

- {1,cos⁡nx,sin⁡nx}n=−∞∞\{1, \cos nx, \sin nx\}_{n=-\infty}^\infty
- {1,cos⁡nx,sin⁡nx}n=0∞\{1, \cos nx, \sin nx\}_{n=0}^\infty
- {1,cos⁡nx,sin⁡nx}n=1∞\{1, \cos nx, \sin nx\}_{n=1}^\infty ✓
- {1,cos⁡nx,sin⁡nx}n=−∞0\{1, \cos nx, \sin nx\}_{n=-\infty}^0

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**7. A given function f(x)f(x) can be represented as the sum of a convergent power series in xx, if f(x)f(x) must have (be) in the neighborhood of x=0x = 0.**

- Continuous
- Extreme points

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- Derivatives of finite order
- Derivatives of all order ✓

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**8. The Fourier series coefficient aka_k is expressed as _____.**

- ak=1π∫−ππf(x)cos⁡(kx)dxa_k = \frac{1}{\pi} \int_{-\pi}^\pi f(x) \cos(kx) dx ✓
- ak=1π∫0πf(x)cos⁡(kx)dxa_k = \frac{1}{\pi} \int_0^\pi f(x) \cos(kx) dx
- ak=1π∫−ππsin⁡(kx)cos⁡(kx)dxa_k = \frac{1}{\pi} \int_{-\pi}^\pi \sin(kx) \cos(kx) dx
- ak=1π∫−ππf(x)sin⁡(kx)dxa_k = \frac{1}{\pi} \int_{-\pi}^\pi f(x) \sin(kx) dx

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**9. If f(z)f(z) and g(z)g(z) are periodic with the same ww, then h(x)=af(zx)+bg(z)h(x) = af(zx) + bg(z) is also a periodic function with period ww.**

- Different period
- Sine
- Cosine
- Same period ✓

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